

Week 3 Interim Submission

Objective

Map the current Emergency Department (ED) triage workflow and identify realistic points where an AI-assisted triage support system could integrate without disrupting clinical practice.

Draft Workflow Notes

Observed / Assumed Current Workflow

Step 1 - Patient Arrival

Patient enters the Emergency Department (walk-in or ambulance arrival).

Information captured:

- Arrival time
- Method of arrival
- Immediate visible concerns

Potential delays:

- Queue at registration
- High patient volume

Possible AI support:

- Predict arrival surges using historical patterns
- Digital intake assistance

Step 2 - Registration

Administrative staff collect patient information.

Information captured:

- Demographics
- Insurance/administrative details
- Reason for visit

Decision point:

- Is immediate emergency intervention required?

Potential issues:

- Duplicate information entry
- Long registration times

AI Plug-in Point 1: Intake Information Extraction

- Convert intake forms into structured fields
- Reduce manual entry burden

Step 3 - Vitals Collection

Triage nurse records observations.

Information captured:

- Blood pressure
- Heart rate
- Respiratory rate
- Temperature
- Oxygen saturation
- Pain score

Decision point:

- Are vitals clinically abnormal?

Potential issues:

- Delayed documentation
- Missing observations

AI Plug-in Point 2: Abnormal Vitals Detection

- Flag unusual combinations of vital signs
- Highlight potential high-risk cases

Step 4 - Triage Nurse Assessment

Clinical assessment performed.

Information captured:

- Chief complaint
- Medical history
- Clinical observations
- Risk screening

Decision point:

- Assign urgency category

Potential issues:

- Subjective variation
- Time pressure

AI Plug-in Point 3: Triage Decision Support

- Suggest urgency category
- Provide risk stratification
- Surface relevant historical patterns

Step 5 - Priority Assignment & Waiting Area

Patient waits according to urgency.

Decision point:

- Should patient be escalated?

Potential issues:

- Queue bottlenecks
- Dynamic patient deterioration

AI Plug-in Point 4: Wait-Time Prediction

- Estimate delays
- Flag prolonged waits

Step 6 - Clinician Assessment

Doctor evaluates patient and determines care.

Activities:

- Examination
- Investigations
- Treatment planning

AI Plug-in Point 5: Disposition Support

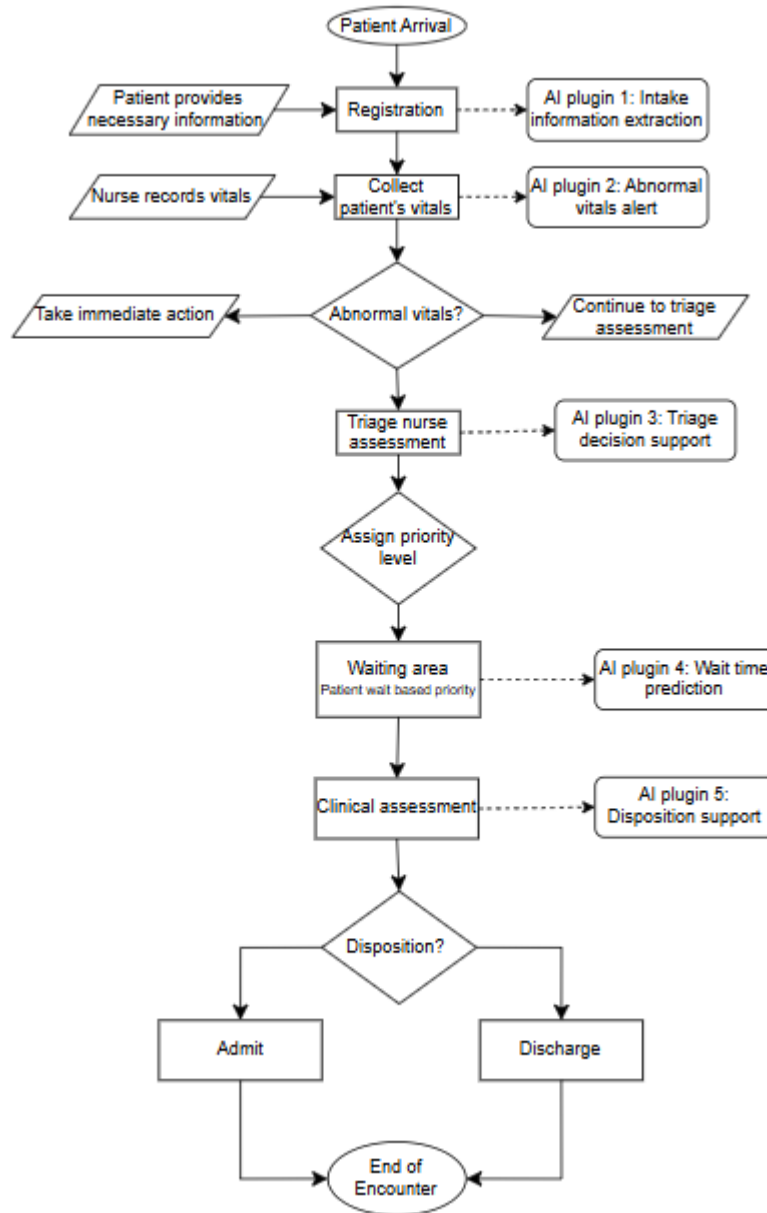
- Predict admission likelihood
- Highlight similar cases

Step 7 - Disposition

Outcome:

- Discharge
- Admission

Initial Workflow Diagram



Initial Observations

- Information enters the system at multiple points rather than once.
- Clinical decisions rely on incomplete and evolving information.
- Multiple handoffs increase risk of information loss.
- Any AI solution must support workflow rather than introduce additional documentation burden.

Candidate Workflow Constraints

Purpose

To identify operational constraints within the Emergency Department (ED) workflow that an AI-assisted triage system must respect in order to create clinical value and support existing processes.

Constraint 1: Limited Nurse Time and Documentation Burden

Emergency Department triage nurses operate in a high-pressure environment where patients arrive continuously and decisions must be made quickly. Additional documentation requirements can slow triage throughput.

If an AI system requires clinicians to manually enter extra fields, open another application, or perform duplicate documentation, adoption is likely to be low.

The AI solution should:

- integrate into existing workflow tools where possible
- minimize clicks and duplicate data entry
- provide outputs within seconds

Constraint 2: Data Quality and Information Lag

Patient information may be incomplete, entered late, captured on paper first, or recorded inconsistently across staff and shifts.

Examples:

- missing vital signs
- delayed documentation
- incomplete chief complaint descriptions

AI performance depends heavily on timely and accurate inputs. Missing or delayed data may reduce reliability and lead to inappropriate recommendations.

The AI system should:

- tolerate missing values
- clearly indicate confidence levels
- avoid producing recommendations from insufficient information

Constraint 3: Clinical Trust and Workflow Adoption

Clinical staff may resist systems that interrupt established workflows or provide recommendations without explanation.

Even accurate systems fail if clinicians do not trust or use them.

The AI should:

- act as decision support rather than decision replacement
- explain contributing factors
- preserve clinician autonomy

Initial Reflection

These constraints suggest that successful healthcare AI depends not only on predictive performance but also on integration into existing clinical workflows, trust, and operational feasibility.